IGCSE 0607 Practice Questions — E5.7

Circle theorems II

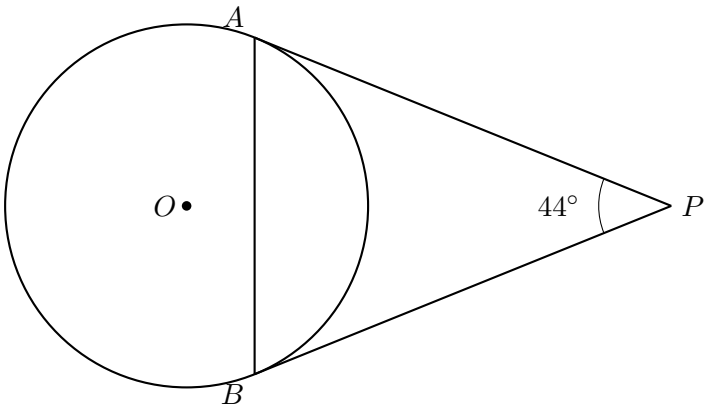
Total Questions: 24 — **Total Marks:** 156 Marks

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- **E5.7** Circle theorems II
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*

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1 The diagram shows a circle with centre O.
P is an external point.
The lines PA and PB are tangents to the circle at the points A and B.
Angle $APB = 44°$.



NOT TO SCALE

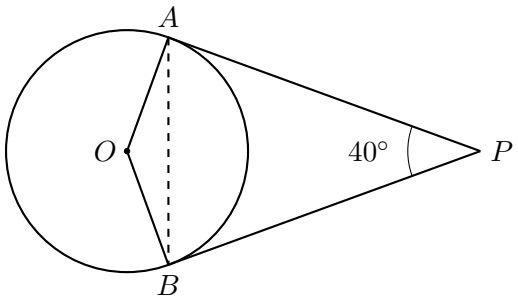
(a) Calculate angle PAB.

Angle PAB = ... ° **[2]**

(b) Calculate angle AOB.

Angle AOB = ... ° **[2]**

2 P is a point outside a circle, centre O.
PA and PB are tangents to the circle at A and B.
Angle $APB = 40°$.



(a) Write down the geometrical reason why triangle PAB is isosceles.

..... **[1]**

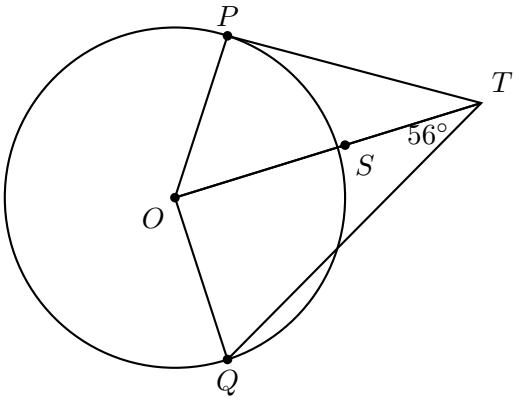
(b) Calculate angle PAB.

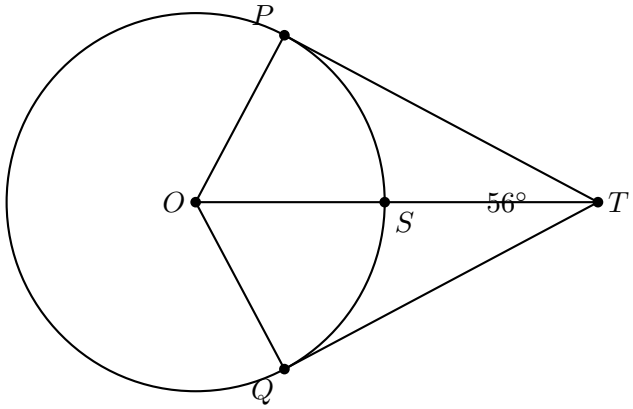
Angle $PAB =$ ..... ° **[1]**

(c) Calculate angle AOB.

Angle $AOB =$ ..... ° **[2]**

3 P and Q are points on a circle, centre O.
TP and TQ are tangents to the circle from an external point T.
The line TO intersects the circle at S.
Angle $PTQ = 56°$.





(a) Calculate angle POT.

Angle $POT =$ ..... ° **[2]**

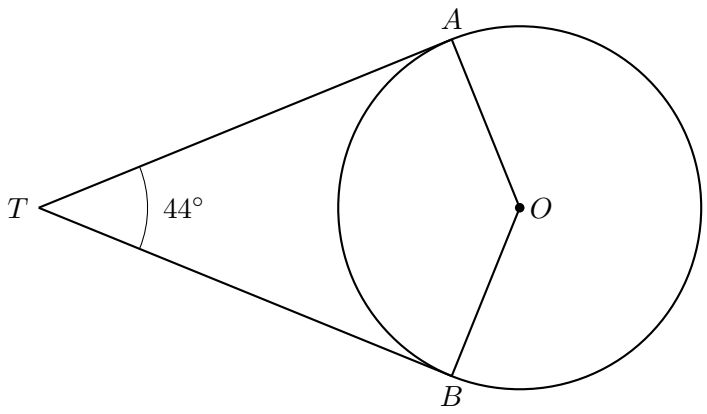
(b) Explain why triangle TPQ is isosceles.

..... **[1]**

(c) Calculate angle TPQ.

Angle $TPQ =$ ..... ° **[1]**

4 The diagram shows a circle, centre O.
TA and TB are tangents to the circle at A and B.
Angle $ATB = 44°$.



(a) Work out angle AOB.

Angle $AOB =$ ..... ° **[2]**

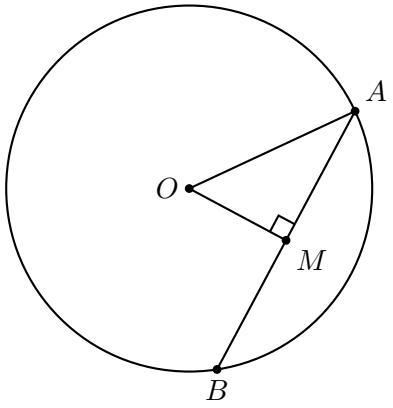
(b) Work out angle OAB.

Angle $OAB =$ ..... ° **[2]**

(c) Write down the geometrical reason why the length of TA is equal to the length of TB.

..... **[1]**

5 The diagram shows a circle, centre O, with radius 10 cm.
AB is a chord of length 16 cm and M is the midpoint of AB.



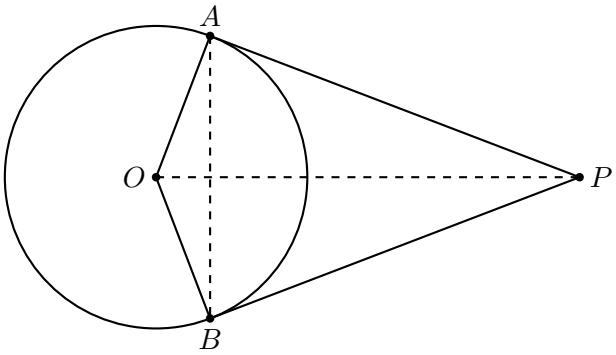
(a) State the size of angle OMA, giving a geometrical reason for your answer.

$\angle OMA =$ ..... ° **[2]**

(b) Calculate the distance OM of the chord AB from the centre O.

$OM =$ ..... cm **[3]**

6 P is an external point.
PA and PB are tangents to a circle, centre O.
The line OP intersects the chord AB at right angles.
$OA = 7$ cm and $PA = 24$ cm.



(a) Write down the length of PB.

..... cm **[1]**

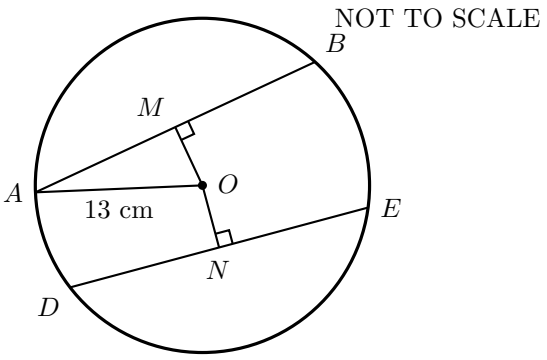
(b) Calculate the length of OP.

..... cm **[2]**

(c) Calculate the shortest distance from O to the chord AB.

..... cm **[3]**

7 The diagram shows a circle, centre O, of radius 13 cm. Chords AB and DE each have length 24 cm. The line OM is perpendicular to AB at M, and the line ON is perpendicular to DE at N.



(a) Write down the length of AM and give a geometrical reason for your answer.

$AM =$ ..... , Reason: ..... **[2]**

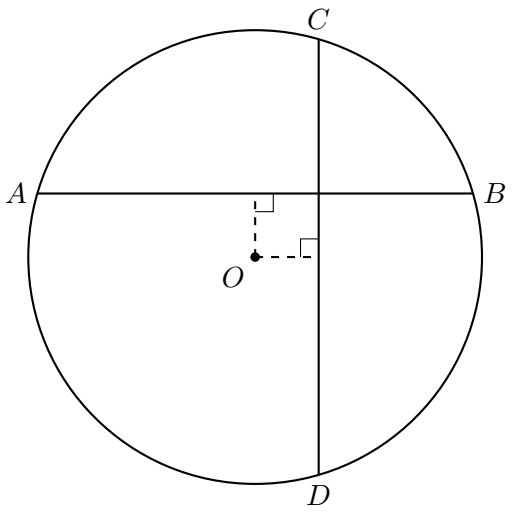
(b) Calculate the distance OM.

$OM =$ ..... cm **[2]**

(c) Write down the length of ON and give a geometrical reason for your answer.

$ON =$ ..... , Reason: ..... **[2]**

8 The diagram shows a circle, centre O, with radius 25 cm.
AB and CD are two chords of the circle such that $AB = CD$.
The perpendicular distance from O to AB is $(x + 3)$ cm.
The perpendicular distance from O to CD is $(3x - 5)$ cm.



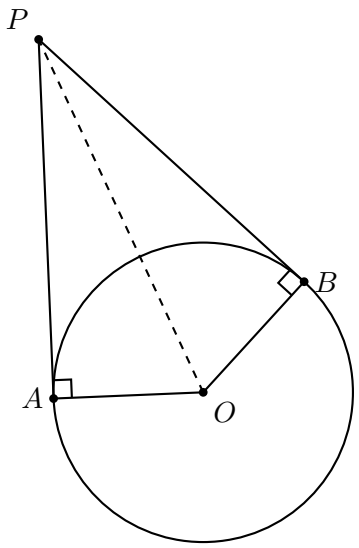
(a) Find the value of x.

$x =$ ..... **[2]**

(b) Calculate the length of the chord AB.

$AB =$ ..... cm **[3]**

9 The diagram shows a circle, centre O, with radius 5 cm.
PA and PB are tangents to the circle at A and B respectively.
The length of PA is 12 cm.



(a) Write down the length of PB, giving a geometrical reason for your answer.

$PB =$ ..... cm **[2]**

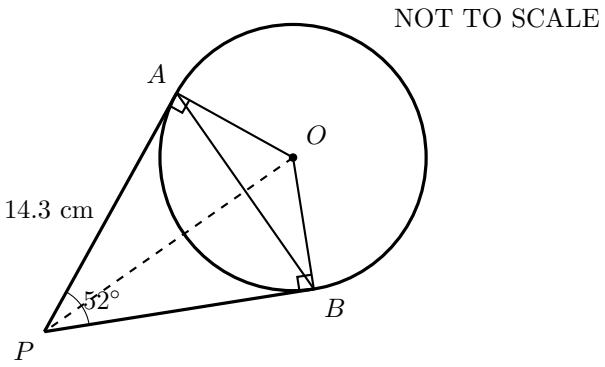
(b) Calculate the distance OP.

$OP =$ ..... cm **[2]**

(c) Calculate the area of the quadrilateral $OAPB$.

Area = ... cm^2 **[2]**

10 The diagram shows a circle, centre O, and an external point P.
The straight lines PA and PB are tangents to the circle at A and B respectively.
Angle $APB = 52°$ and $PA = 14.3$ cm.



(a) Explain why triangle OAP is congruent to triangle OBP.

.....

.....

..... **[3]**

(b) Calculate the size of

(i) angle AOB,

Angle $AOB =$ ..... ° **[2]**

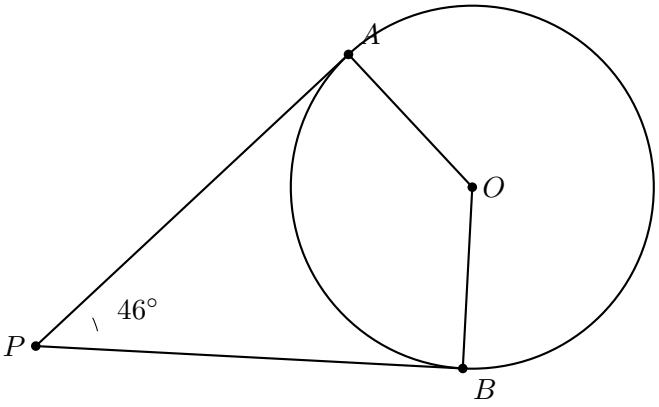
(ii) angle OAB.

Angle $OAB =$ ..... ° **[2]**

(c) Write down the length of PB.

$PB =$ ..... cm **[1]**

11 P is an external point.
PA and PB are tangents to the circle, centre O, at the points A and B.
Angle $APB = 46°$.



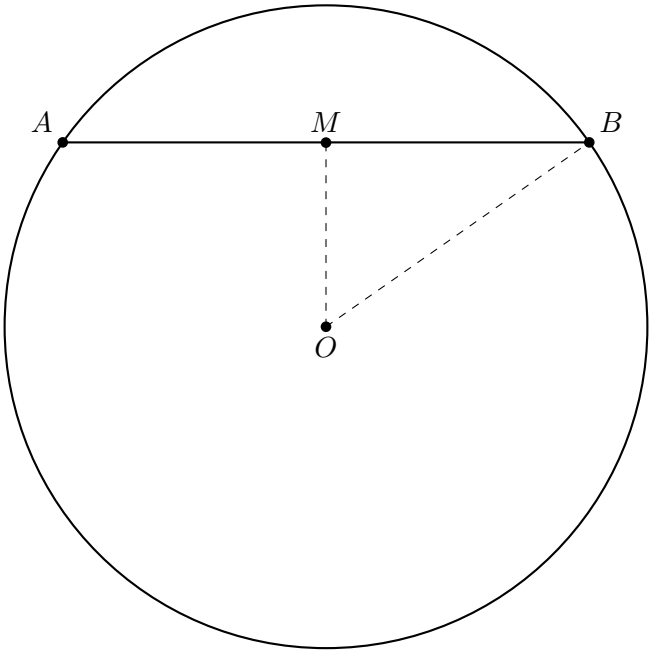
(a) Calculate angle AOB.

Angle $AOB =$ ..... ° **[2]**

(b) Calculate angle OAB.

Angle $OAB =$ ..... ° **[2]**

12 A circle, centre O, has a radius of 8.5 cm.
A chord AB has length 14 cm.
M is the midpoint of AB.



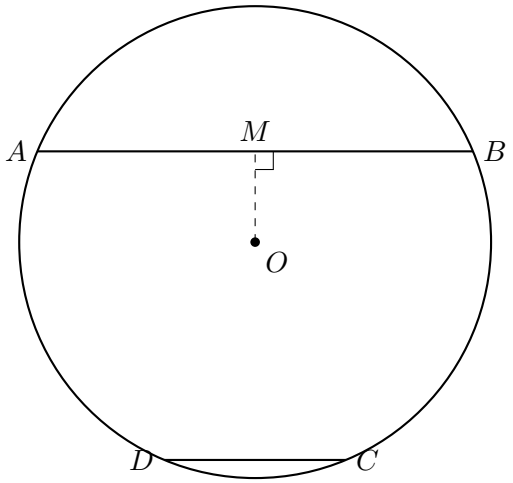
(a) Calculate the distance OM.

$OM =$ ..... cm **[3]**

(b) Another chord CD is drawn such that it is parallel to AB and is at a distance of 3 cm from O.
Calculate the length of the chord CD.

$CD =$ ..... cm **[2]**

13 The diagram shows a circle, centre O, with radius 13 cm.
AB is a chord of the circle of length 24 cm.
The point M is the midpoint of the chord AB.
Another chord CD is drawn inside the circle.



NOT TO SCALE

(a) Calculate the perpendicular distance, OM, from the centre of the circle to the chord AB.

$OM =$ ..... cm **[3]**

(b) The perpendicular distance from the centre O to the chord CD is also 5 cm.
Write down the length of the chord CD.

$CD =$ ..... cm **[1]**

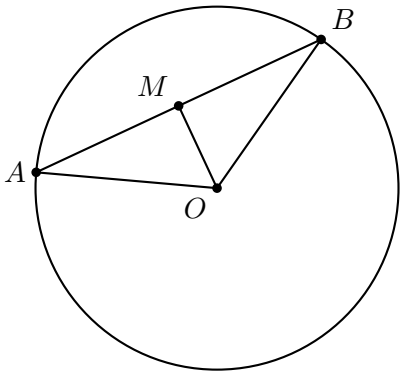
(c) Calculate the area of triangle OAB.

Area =

.....

cm^2 **[2]**

14 A and B are points on a circle, centre O and radius 13 cm.
The length of the chord AB is 24 cm.
The point M is the midpoint of AB.



(a) State the geometrical reason why angle $OMA = 90°$.

..... **[1]**

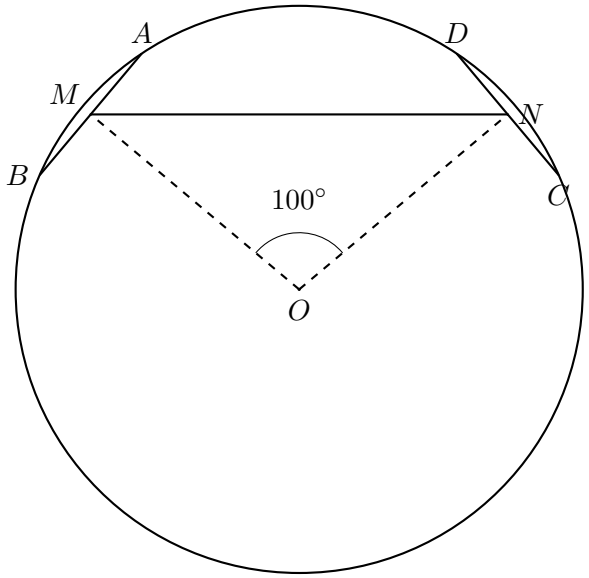
(b) Calculate the length of OM.

$OM =$ ..... cm **[3]**

(c) Calculate angle AOB.

Angle $AOB =$ ..... ° **[3]**

15 A, B, C and D are points on a circle, centre O and radius 25 cm.
The chords AB and CD each have a length of 14 cm.
M is the midpoint of AB and N is the midpoint of CD.
Angle $MON = 100°$.



(a) Show that $OM = 24$ cm.

[2]

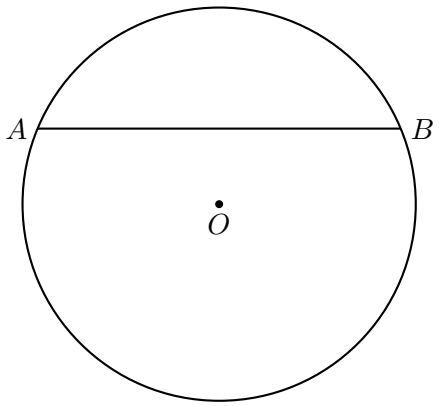
(b) Calculate the length of MN.

..... cm **[3]**

(c) Calculate the area of triangle MON.

..... cm^2 **[2]**

16 The diagram shows a circle, centre O, with radius 13 cm.
AB is a chord of the circle of length 24 cm.



(a) Calculate the shortest distance from O to the chord AB.

..... *cm* **[3]**

(b) CD is another chord of the circle.
CD is parallel to AB and has length 24 cm.
Write down the shortest distance from O to CD.

..... *cm* **[1]**

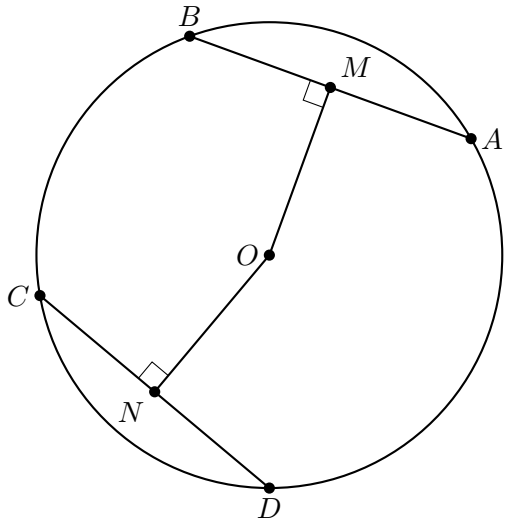
(c) EF is a third chord of the circle.
The shortest distance from O to EF is 12 cm.
Calculate the length of the chord EF.

..... *cm* **[3]**

(d) State the geometrical property of a circle used to find the length of half the chord in part (a).

..... **[1]**

17 AB and CD are two chords of a circle, centre O.
The chords AB and CD are equal in length.
The line OM is the perpendicular bisector of AB, and the line ON is the perpendicular bisector of CD.
$OM = (5x - 7)$ cm and $ON = (2x + 2)$ cm.
The radius of the circle is 17 cm.



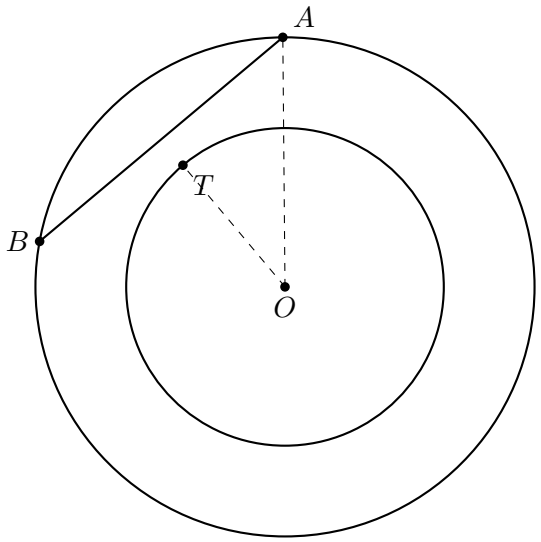
(a) Find the value of x.

$x =$ ..... **[3]**

(b) Calculate the length of the chord AB.

$AB =$ ..... cm **[4]**

18 Two circles, C_1 and C_2, have a common centre O.
The radius of C_1 is r cm and the radius of C_2 is R cm, where $R > r$.
A chord AB of the larger circle C_2 is tangent to the smaller circle C_1 at the point T.
The length of the chord AB is 24 cm and the area of the annular region between the two circles is $k\pi$ cm^2.



(a) Show that $R^2 - r^2 = 144$.

[3]

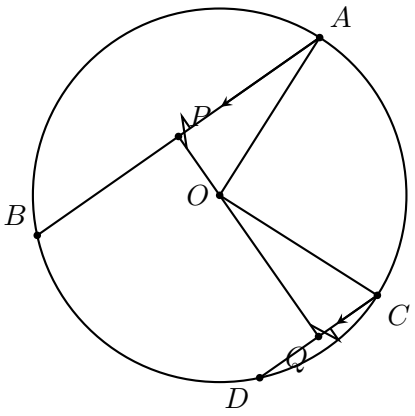
(b) Find the value of k.

$k =$ ..... **[2]**

(c) A second chord CD of the larger circle C_2 has length 24 cm.
State, with a reason, the distance from O to the chord CD.

..... **[2]**

19 The diagram shows a circle, centre O, with two parallel chords AB and CD on opposite sides of the centre.
$AB = 24$ cm, $CD = 10$ cm, and the perpendicular distance between the two chords is 17 cm.
P and Q are the midpoints of AB and CD respectively.



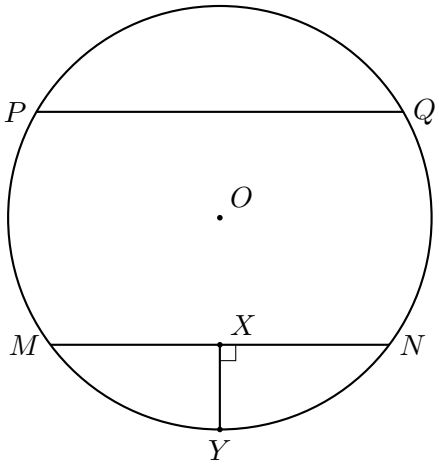
(a) Explain why P, O and Q lie on a straight line.

..... **[2]**

(b) Let $OP = x$ cm and the radius of the circle be r cm.
Calculate the radius, r, of the circle.

$r =$ ..... cm **[5]**

20 A circle has centre O and radius 10 cm.
MN is a chord of length 16 cm.
The perpendicular bisector of MN intersects MN at X and the minor arc MN at Y.
PQ is a second chord which is parallel to MN.
The perpendicular distance between MN and PQ is 11 cm.



(a) Explain why the line containing XY passes through the centre O.

..... **[1]**

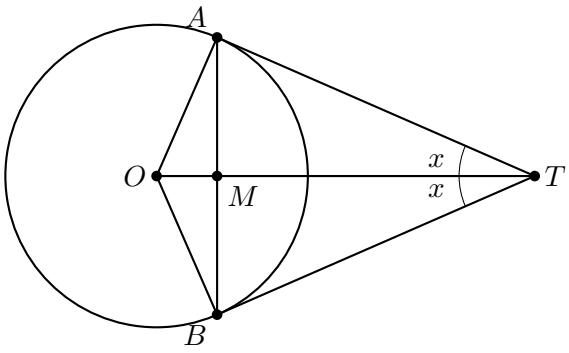
(b) Calculate the length of XY.

$XY =$ ..... cm **[3]**

(c) Calculate the length of the chord PQ.

$PQ =$ ..... cm **[4]**

21 TA and TB are tangents to a circle, centre O and radius r, at the points A and B.
The line TO intersects the chord AB at M.
Angle ATO = Angle $BTO = x$.



(a) Show that the length of the chord AB is $2r\cos x$.

[2]

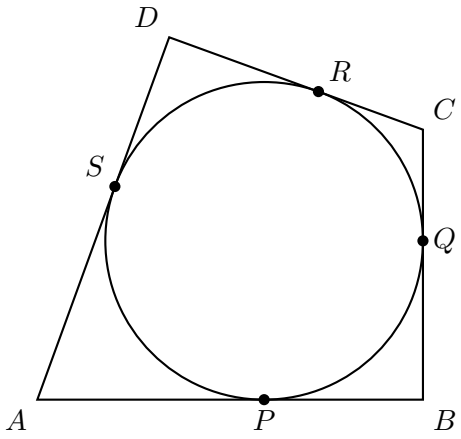
(b) A second chord BC is drawn such that $BC = AB$.
Write down the shortest distance from O to the chord BC, in terms of r and x.

..... **[1]**

(c) The area of triangle OAB is equal to the area of triangle ATB.
Find the exact value of x.

$x =$ ..... $^\circ$ **[5]**

22 A circle is inscribed in a quadrilateral $ABCD$.
The circle touches the sides AB, BC, CD, and DA at the points P, Q, R, and S respectively.



(a) Write down the geometrical reason why $AP = AS$.

..... **[1]**

(b) Show that $AB + CD = AD + BC$.

[3]

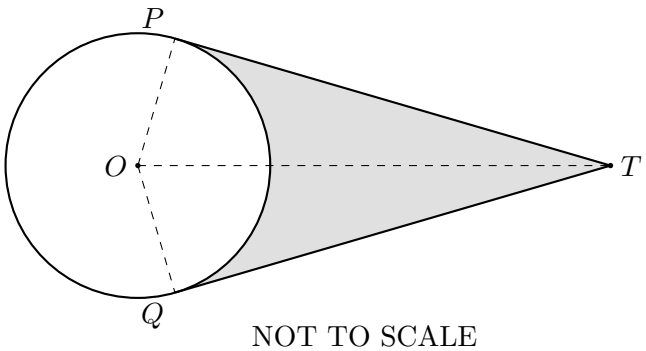
(c) Given that $AB = (2x + 5)$ cm, $BC = (3x - 2)$ cm, $CD = (x + 8)$ cm and $AD = (2x + 1)$ cm, work out the value of x.

$x =$ ..... **[3]**

(d) Calculate the perimeter of the quadrilateral $ABCD$.

Perimeter = ... cm **[2]**

23 The diagram shows a circle with centre O and radius 7 cm.
The lines TP and TQ are tangents to the circle at the points P and Q.
The distance from the centre O to the external point T is 25 cm.



NOT TO SCALE

(a) Show that the length of the tangent TP is 24 cm.

[2]

(b) Calculate angle PTQ.

Angle PTQ = ... ° **[3]**

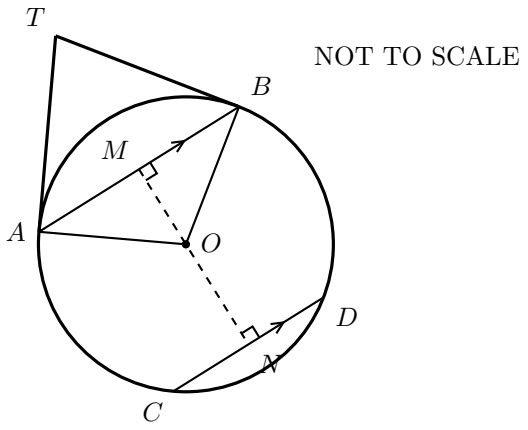
(c) Calculate the area of the shaded region bounded by the tangents TP, TQ and the minor arc PQ.

Area =

.....

cm^2 **[4]**

24 The diagram shows a circle, centre O, of radius r cm. Chords AB and CD are parallel and lie on opposite sides of the centre O.
$AB = 16$ cm, $CD = 12$ cm, and the perpendicular distance between AB and CD is 14 cm. The tangents to the circle at A and B meet at the external point T.



(a) The perpendicular line through O meets AB at M and CD at N, with $OM = x$ cm. Write down two equations in terms of r and x, and solve them to show that $x = 6$ and $r = 10$.

[4]

(b) Calculate the exact length of tangent TA, giving your answer as an improper fraction.

$TA =$ ..... cm **[3]**

(c) Write down the length of TB, and calculate the perimeter of quadrilateral $OATB$, giving your answer correct to 1 decimal place.

$TB =$ ..... , Perimeter = ... **[2]**

(d) Calculate the size of angle ATB, giving your answer correct to 1 decimal place.

Angle $ATB =$ ..... ° **[2]**